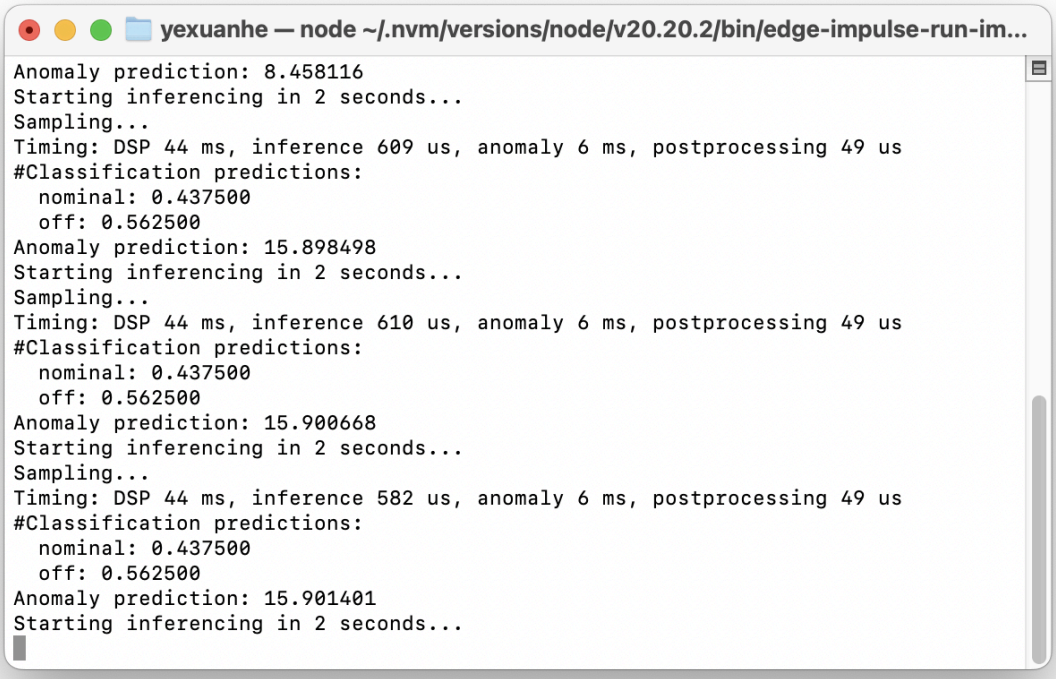
Part 1

1. Raw inputs: State the raw inputs used in the Edge Impulse project.
    - time series data: accX, accY, accZ
2. NN classifier features: State the type of features passed into the NN classifier, the number of input features, and at least five actual feature names used to train the classifier.
    - Spectral features
    - 33 features
    - accX RMS, accX Peak 1 Height, accY Peak 3 Height, accX Peak 1 Freq, accY RMS
3. NN classifier outputs: State the outputs of the NN classifier.
    - Nominal / off
4. Anomaly detector features: State the type of features passed into the anomaly detector, the number of features used, and the feature names used to train the anomaly detector.
    - Spectral features
    - 3 features
    - accX Peak 1 Height, accY Spectral Power 2.0 - 5.0 Hz, accZ Spectral Power 2.0 - 5.0 Hz
5. Deployment evidence: Include a screenshot showing the Edge Impulse model running on the Arduino Nano 33 BLE Sense and producing inference results.

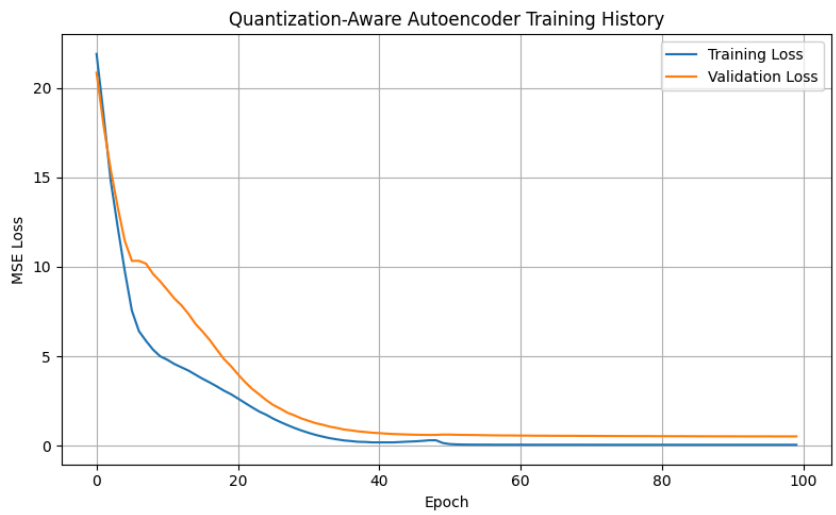


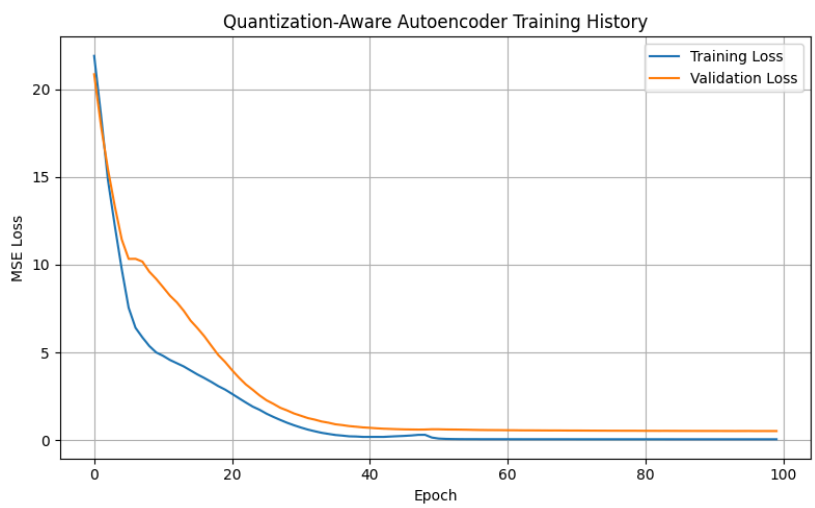
6. Short interpretation: Briefly interpret the observed deployment results. Discuss whether the classifier output should always be trusted and under what conditions it may or may not be reliable.

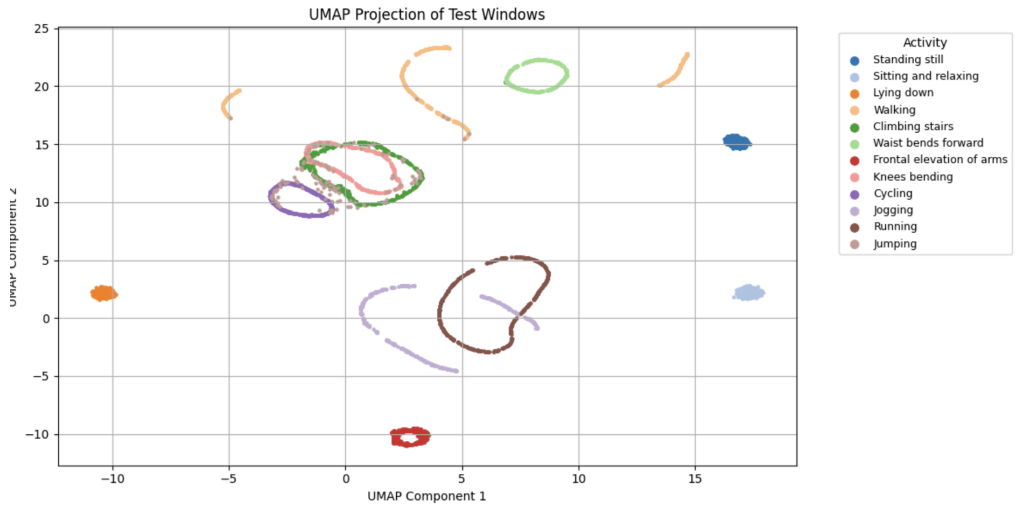
- During deployment, the model repeatedly predicted the fan as off with about 56% confidence, while nominal was about 44%. This suggests the fan may be off, but the result should not be fully trusted because the confidence is low and the prediction is close. The classifier is more reliable when the sensor is placed in the same way as during training, and the prediction confidence is high. It is less reliable if the sensor moves, the fan speed changes, there is outside vibration, or the prediction scores are close together, like this.

Part 2

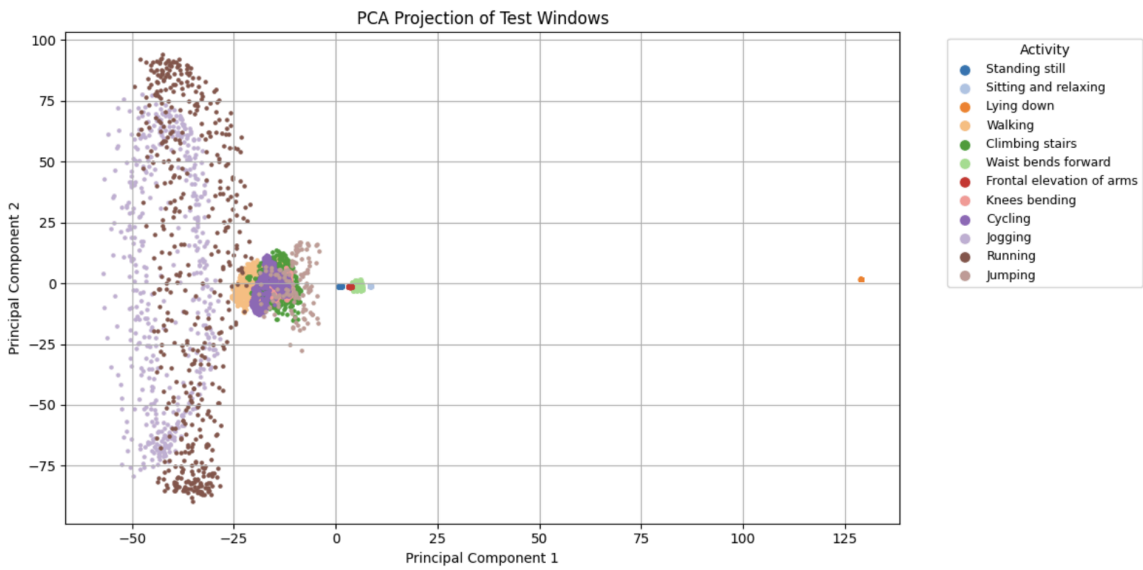
Training and validation loss curves



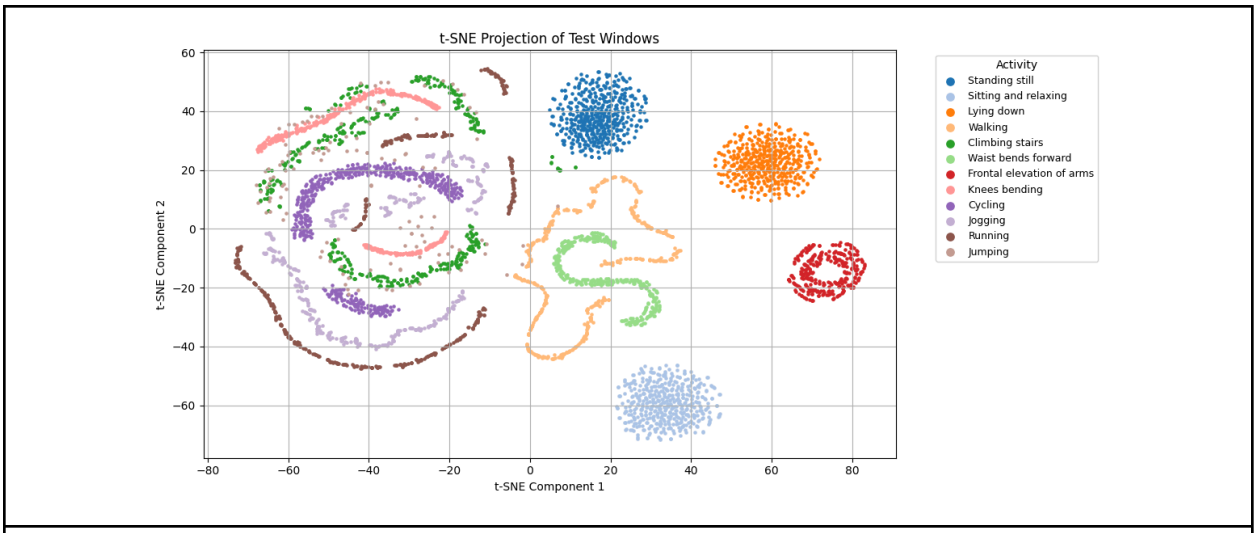



UMAP Projection of Test Windows
UMAP Component 1
UMAP Component 2
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping

UMAP


PCA Projection of Test Windows
Principal Component 1
Principal Component 2
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping

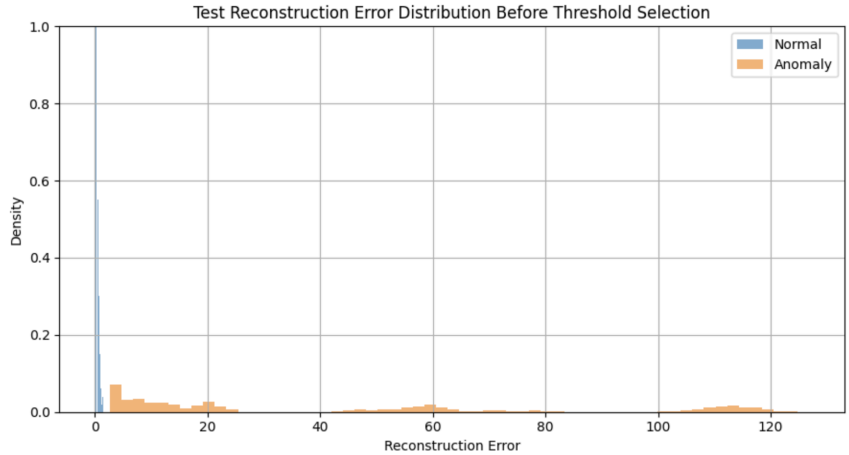
PCA

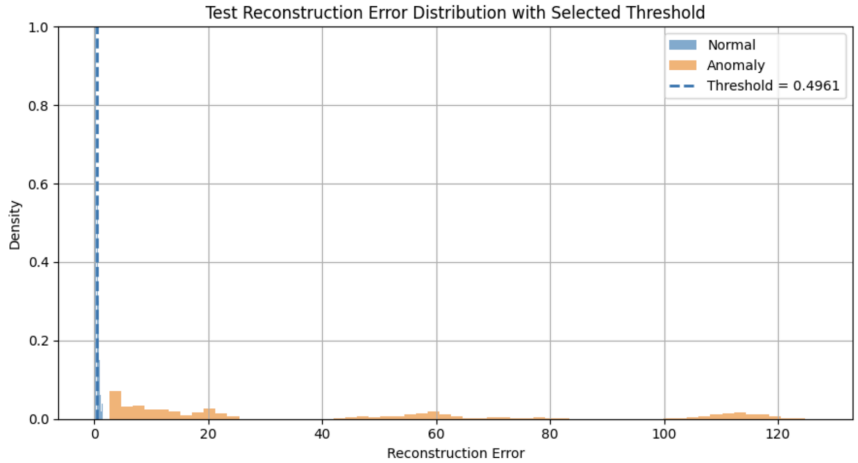


t-SNE

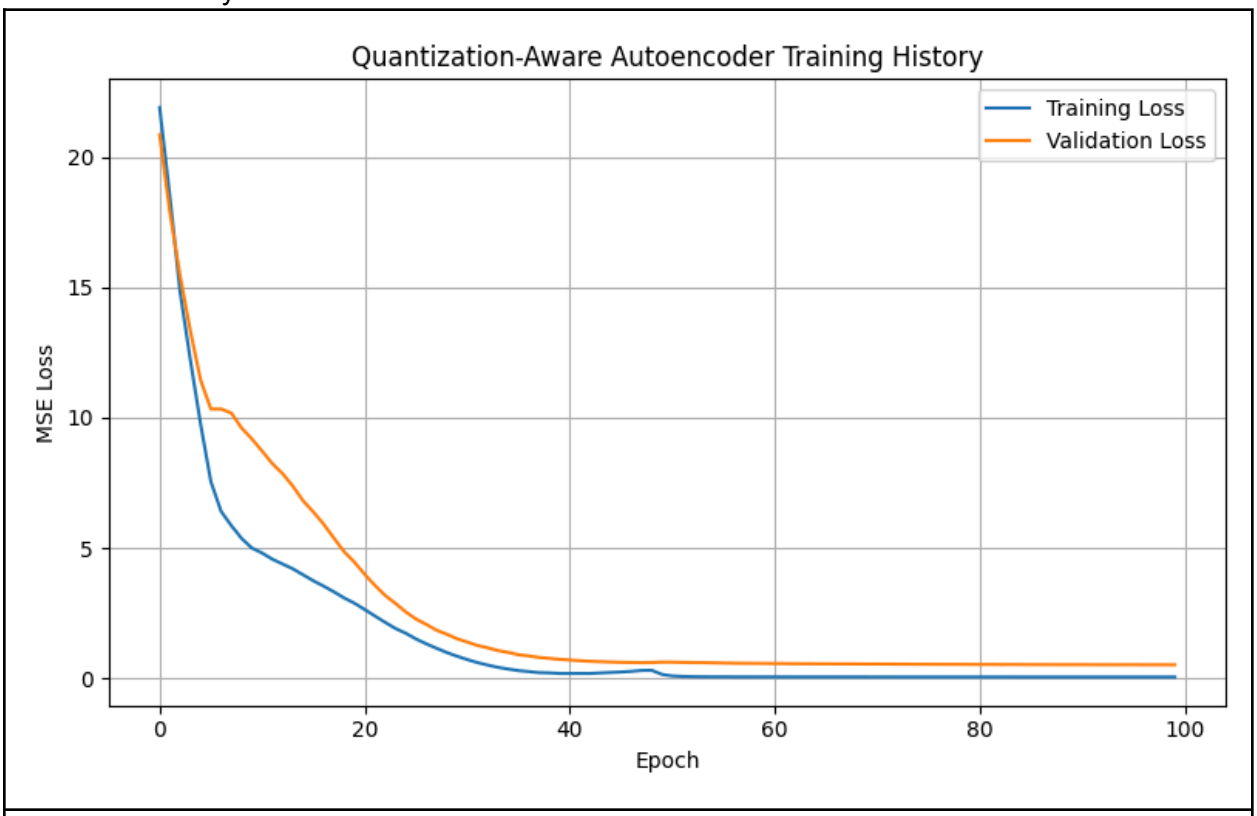
Reconstruction error distribution

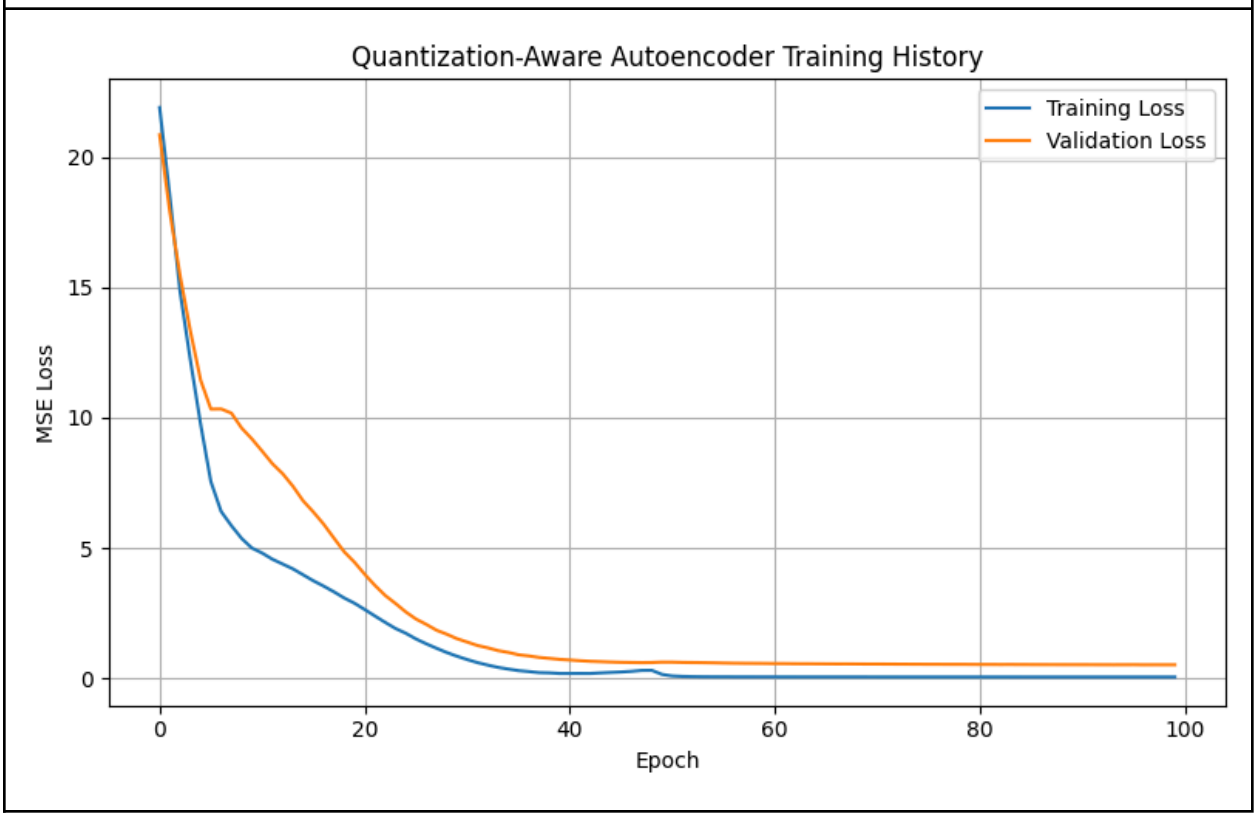
```
Train normal       mean=0.095231, stderr=0.001656, min=0.007955, max=1.089465
Train anomaly      mean=41.514839, stderr=0.366109, min=2.302257, max=124.967941
Test normal        mean=0.135468, stderr=0.004005, min=0.008792, max=1.448806
Test anomaly       mean=41.902416, stderr=0.614349, min=2.634231, max=126.787903
```



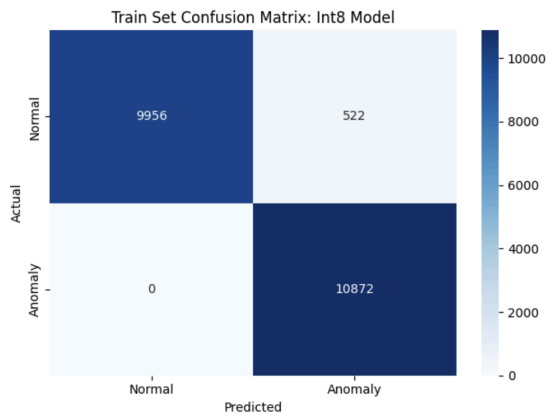


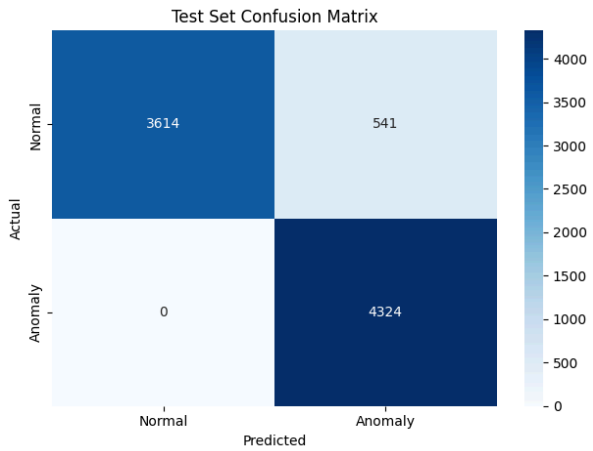
Selected anomaly threshold





```
Train set evaluation
              precision    recall  f1-score   support

  Normal (0)       1.00      0.95      0.97     10478
 Anomaly (1)       0.95      1.00      0.98     10872

    accuracy                           0.98     21350
   macro avg       0.98      0.97      0.98     21350
weighted avg       0.98      0.98      0.98     21350

Test set evaluation
              precision    recall  f1-score   support

  Normal (0)       1.00      0.87      0.93      4155
 Anomaly (1)       0.89      1.00      0.94      4324

    accuracy                           0.94      8479
   macro avg       0.94      0.93      0.94      8479
weighted avg       0.94      0.94      0.94      8479
```





```
Reconstruction error: 0.249633
Result: Normal activity
Reconstruction error: 0.249618
Result: Normal activity
Reconstruction error: 0.249603
Result: Normal activity
Reconstruction error: 0.249595
Result: Normal activity
Reconstruction error: 0.249574
Result: Normal activity
Reconstruction error: 0.249566
Result: Normal activity
Reconstruction error: 0.249558
Result: Normal activity
Reconstruction error: 0.249549
Result: Normal activity
Reconstruction error: 0.249538
Result: Normal activity
Reconstruction error: 0.249518
Result: Normal activity
Reconstruction error: 0.249497
Result: Normal activity
Reconstruction error: 0.249501
Result: Normal activity
Reconstruction error: 0.249484
Result: Normal activity
Reconstruction error: 0.249479
Result: Normal activity
Reconstruction error: 0.249450
Result: Normal activity
```